

Piecewise Linear Regression

Optimization problem:

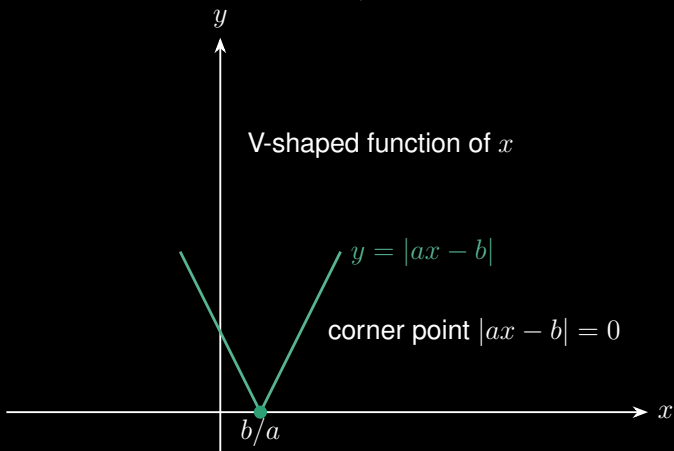
$$\min_x \quad f(x) = \sum_{i=1}^n |a_i x - b_i|$$

Equivalent linear programming:

$$\begin{aligned} \min_{x, t_1, \dots, t_n} \quad & \sum_{i=1}^n t_i \\ \text{s.t.} \quad & -t_i \leq a_i x - b_i \leq t_i, \quad \forall i \end{aligned}$$

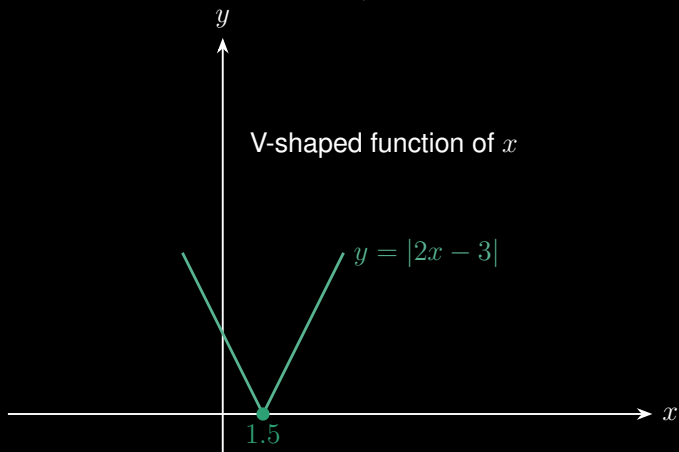
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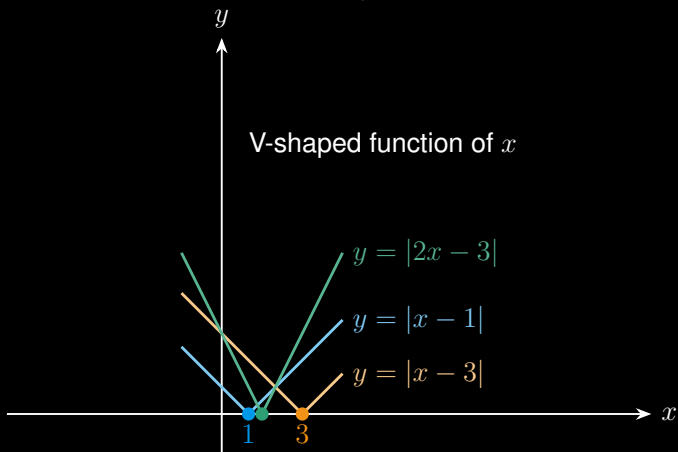
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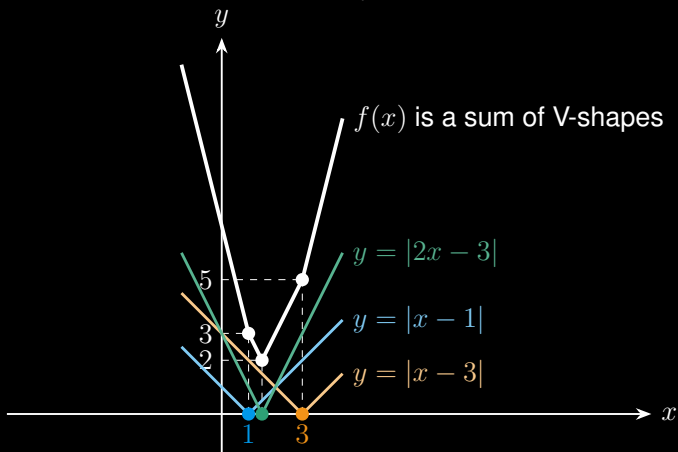
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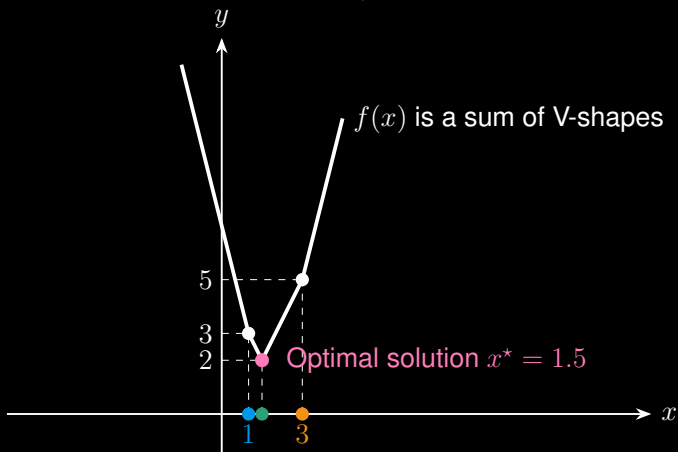
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Thanks for your attention!

About me:

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